

Fama French and Momentum Factors: Indian Market

Project Proposal

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Abstract

A portfolio or risk team starts each day with the same question: what are our holdings really exposed to? Two stocks from different industries can rise and fall together because they share an exposure to size, value or momentum, and a portfolio built from them carries more risk than it appears. This project builds a daily panel of 46 NIFTY 50 stocks from 2020 to 2025, joined with the IIM Ahmedabad Fama-French and momentum factors, and uses it to explore how the factors and the stocks behaved. Even in 2020, the market explained less than half of the average stock's daily movement. The next stage is an interactive R Shiny dashboard that shows the factor exposure of any stock or portfolio, the kind of analysis usually found in expensive institutional risk systems.

1 Introduction and Problem Statement

A stock's daily return is not driven by one thing. Part of it comes from the market as a whole, part from the type of stock it is (small or large, cheap or expensive, recent winner or loser), and part from news about the company itself. For example, on a day Tata Steel falls 3%, part of that may be the whole market falling, part may be value stocks as a group falling more than the rest, and part may be steel-specific news.

Looking only at returns, these parts can't be told apart, and that matters for anyone holding more than one stock. Infosys writes software and Hindustan Unilever sells soap, so holding both looks diversified. But both behave like growth stocks and tend to do badly on the same days, whenever growth stocks lag, so the portfolio is less diversified than it looks.

This project uses the IIM Ahmedabad factors for 46 NIFTY 50 stocks from 2020 to 2025 to show how the factors behaved and how much of each stock's movement came from the market, as a first step towards an interactive dashboard.

2 Data Description

This project combines two data sources for January 2020 to December 2025: daily factor returns from IIM Ahmedabad and daily stock prices from Yahoo Finance, about 83,000 rows of raw data

in total. The factor and price data are downloaded in R, and all code and data are on GitHub: <https://github.com/DhruvDevpura/india-factor-risk-dashboard>.

The four factors: IIM Ahmedabad builds these from BSE and NSE stocks, leaving out micro-cap and penny stocks (Agarwalla, Jacob and Varma, 2013).

- **MF (market):** the return of the whole Indian stock market minus the risk-free rate. It is positive when stocks do better than Treasury Bills.
- **SMB (size, small minus big):** the return of small companies minus the return of large companies. It is positive when small companies do better.
- **HML (value, high minus low):** the return of value stocks, which are cheap compared to their book value, minus the return of growth stocks, which are expensive compared to it. It is positive when value stocks do better.
- **WML (momentum, winners minus losers):** the return of stocks that have recently risen the most minus the return of stocks that have recently fallen the most. It is positive when recent winners keep doing better.

Files used:

- `ff_india_daily_2025_12.csv`: downloaded in R directly from the IIM Ahmedabad data library. 8,004 rows and 6 columns, daily from October 1993 to December 2025, with the date, the four factors and the daily risk-free rate (RF).
- `ind_nifty50list.csv`: the NIFTY 50 constituent list from NSE, as of 14 September 2026, downloaded manually as NSE blocks automated requests. 50 rows and 5 columns, with each company's name, industry and symbol.
- `nse_prices_raw.csv`: daily prices from Yahoo Finance, downloaded in R using the `tq_get()` function from the `tidyquant` package. 74,672 rows and 8 columns, with daily prices for the 50 stocks and the NIFTY 50 index.

Preprocessing steps:

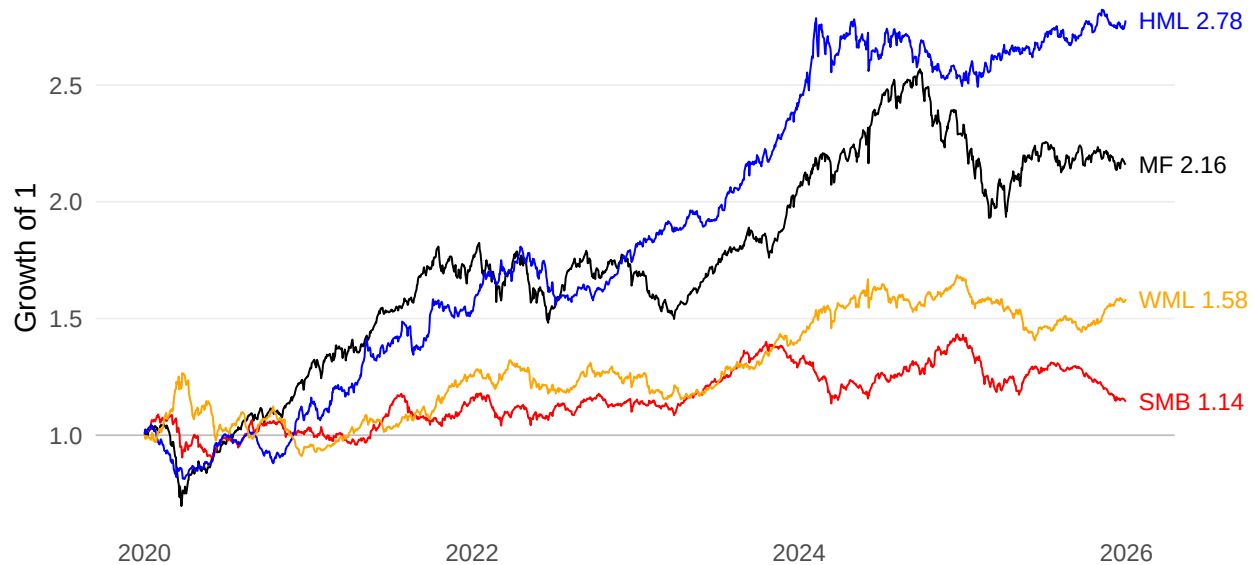
- Converted the IIM Ahmedabad returns from percent to decimals. As a check, RF averages about 5.2% a year over 2020 to 2025, in line with 91-day Treasury Bill yields.
- Dropped four stocks: Eternal, Jio Financial Services and Max Healthcare listed after January 2020, and Tata Motors had a demerger in October 2025 that breaks its price history.
- Computed daily returns for each stock and for the NIFTY 50 index from adjusted closing prices, and subtracted RF from the stock returns to get excess returns.
- Joined everything by date. Seven trading days, mostly special sessions such as Saturday and Diwali Muhurat trading, had data missing in one source and were dropped, leaving 1,485 days with no missing values.

Final panel: 68,310 rows and 10 columns, one row per stock per trading day: the date, the stock's ticker, its daily return (`ret`), the risk-free rate (`rf`), its excess return (`exret`), the four factor returns (`mf`, `smb`, `hml`, `wml`) and the NIFTY 50 index return (`nifty_ret`). Every chart in this proposal is built from this file.

3 Exploratory Data Analysis

3.1 Growth of the four factors

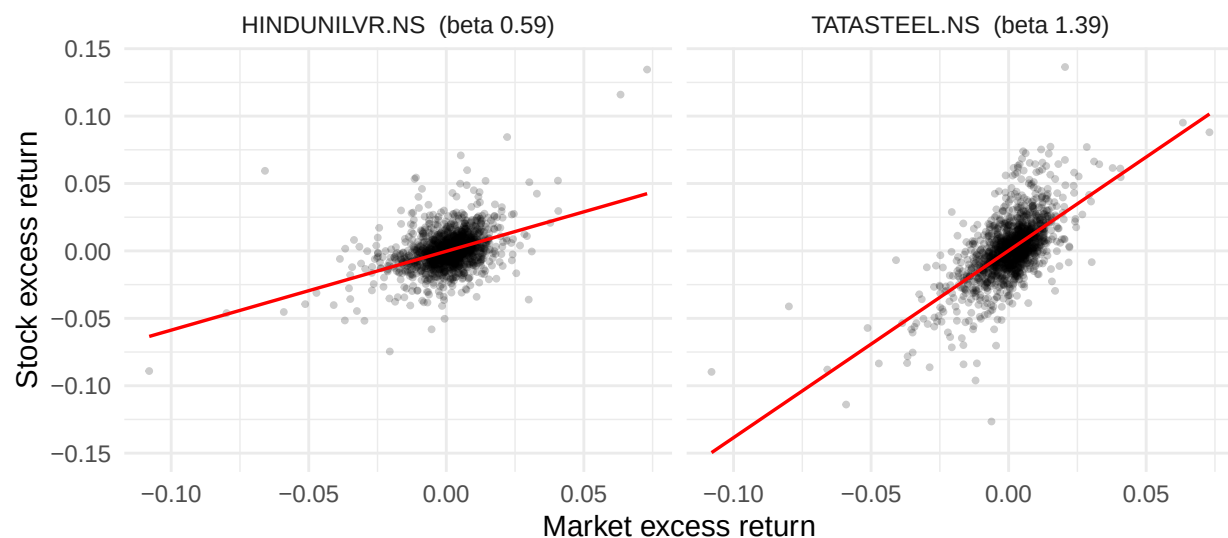
Growth of 1 in each IIM-A factor, 2020 to 2025



Value (HML) did best over the period, growing about 19% a year, and it was the only factor with a positive return in every calendar year. Size (SMB) ended almost where it started. These are not returns an investor could actually earn, since they ignore trading and shorting costs, and MF is the market's return over Treasury Bills rather than the market's own return.

3.2 Market sensitivity of two stocks

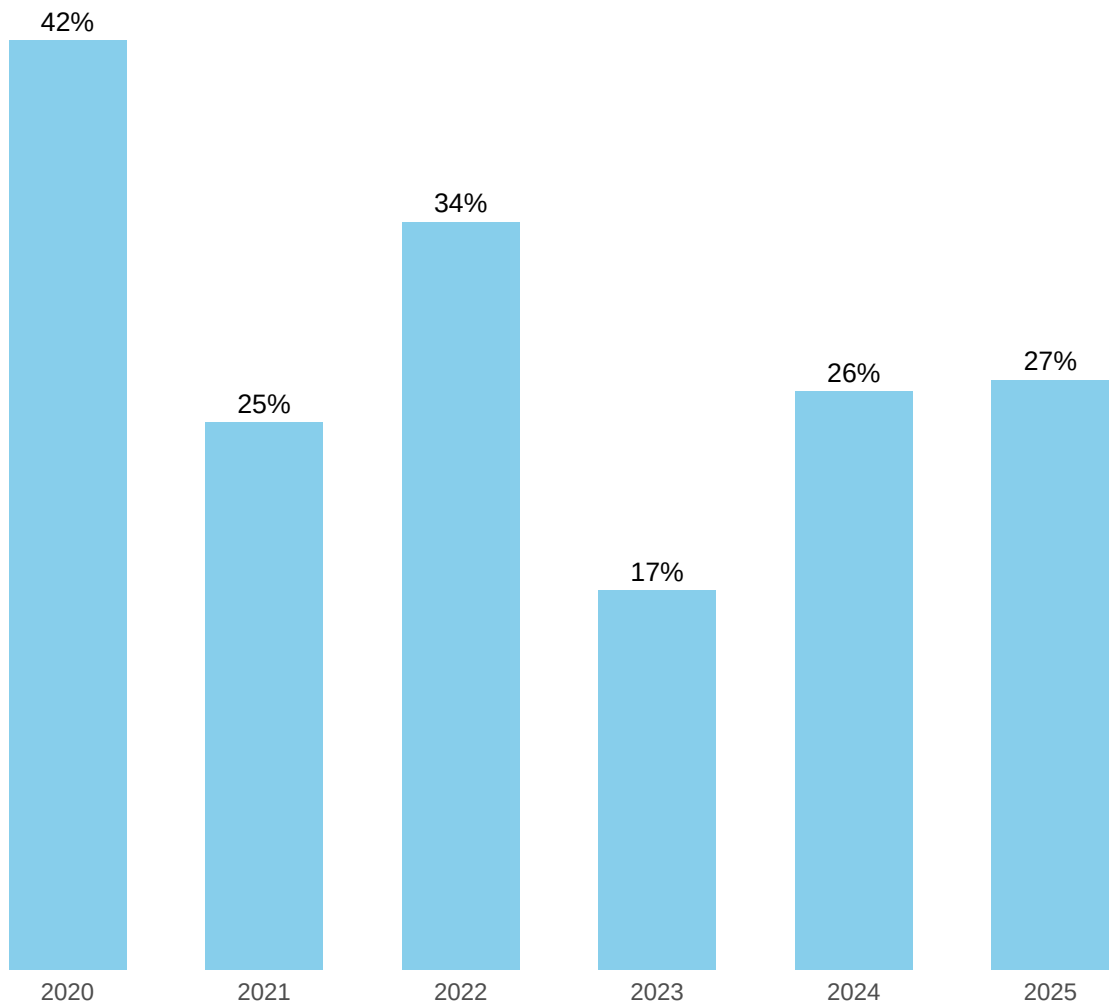
Daily excess returns of HUL and Tata Steel against the market



The slope of each line is the stock’s beta. When the market moves 1%, Hindustan Unilever moves about 0.59% on average and Tata Steel about 1.39%, so Tata Steel falls much harder when the market falls. Tata Steel also follows the market more closely: the market explains about 42% of its daily movement, against 18% for Hindustan Unilever. The two were picked from opposite ends of the 46 stocks.

3.3 Share of movement explained by the market

Share of daily stock movement explained by the market, by year



For each stock and each year, the share is the squared correlation of its daily return with the market, averaged across the 46 stocks. Stocks moved together most in 2020, when the COVID crash hit every company at once, and least in 2023. Even in 2020, the market explained less than half of the average stock’s movement, with the rest coming from the other three factors, industry effects and company news.

3.4 Volatility of the four factors

Annualised volatility of each IIM-A factor, by year

MF	25.7%	14.2%	17.4%	10.3%	15.8%	14.5%
SMB	16.1%	12.6%	12.2%	8.7%	12.2%	11.0%
HML	16.2%	16.4%	12.4%	9.0%	14.9%	7.8%
WML	18.5%	11.0%	12.0%	9.6%	13.6%	8.6%
	2020	2021	2022	2023	2024	2025

Volatility here is the standard deviation of a factor's daily returns, scaled to a year by multiplying by the square root of 252 trading days. The market was the most volatile factor in five of the six years, and 2020 stands out for three of the four factors: the market's typical daily move was about 1.6% that year, against about 0.65% in 2023. A factor's risk is therefore not fixed, and the same exposure to a factor can carry very different risk from one year to the next.

4 Approaches

The next stage turns the panel and charts into an interactive dashboard in R Shiny. Commercial risk models already break portfolios into factor exposures, but they are built and priced for large institutions. This dashboard offers a free and transparent version for Indian stocks, built on IIM Ahmedabad's public data, for smaller investors and advisers. The main pieces are:

- **Sensitivity of each stock to all four factors.** Section 3.2 measures how much a stock moves with the market. The same idea extends to size, value and momentum together, estimated by regressing each stock's excess return on the four factors (Sen and Das, 2023, Chapter 18). This shows what each stock is really a bet on. A first four-factor fit on the panel already shows it: Infosys and Hindustan Unilever have value loadings of -0.58 and -0.54 , so both behave like growth stocks.
- **Splitting each stock's movement into its sources.** Section 3.3 separates the market from everything else. The next step measures how much the other three factors add, leaving

the part that comes from industry and company news. This shows how much of a stock's risk can be reduced by holding more stocks, and how much cannot.

- **Tracking factor risk over time.** Section 3.4 shows that a factor's volatility changes a lot from year to year. The dashboard will recompute it over a moving one-year window, so an investor who wants steady risk can see when a factor has become more volatile and hold less of it.
- **Whether sensitivities change over time.** Each stock's sensitivities are re-estimated one year at a time. This matters to a portfolio manager because decisions are made on past estimates: a stock bought as defensive, with a low beta, may have become more sensitive to the market, and the portfolio would then fall more than planned in a downturn.
- **A portfolio view.** A user picks stocks and weights, and the dashboard shows the portfolio's exposure to each factor as the weighted average of its stocks' sensitivities. This makes hidden concentration visible, such as two holdings that look different but bet on the same factor. It will also split the portfolio's risk into the part explained by the four factors and the part left over, while allowing the leftover parts of different stocks to move together, since stocks from the same industry, such as TCS and Infosys, still share news that the factors do not capture.

5 Conclusion

Taken together, the four charts point the same way: the market alone does not explain these stocks well. The other factors followed very different paths from the market with risk that changed sharply from year to year, stocks carried very different amounts of market risk, and even in 2020 most of the average stock's daily movement came from something other than the market. A view based only on the market would miss much of what drives Indian stocks, which is why the next stage measures all four factors together and how much is left after them.

References and Acknowledgements

1. Agarwalla, S. K., Jacob, J. and Varma, J. R. (2013). Four factor model in Indian equities market. Working Paper No. 2013-09-05, Indian Institute of Management, Ahmedabad. Data library: <https://faculty.iima.ac.in/iffm/Indian-Fama-French-Momentum/>
2. Sen, R. and Das, S. (2023). *Computational Finance with R*. Indian Statistical Institute Series. Springer, Singapore. <https://doi.org/10.1007/978-981-19-2008-0>
3. National Stock Exchange of India. NIFTY 50 constituent list, accessed 14 September 2026.
4. Yahoo Finance. Daily historical stock prices, accessed through the `tidyquant` R package, September 2026.
5. Claude (Anthropic), large language model. Used to write and refine R code, particularly for the charts, and to review drafts. All methodological, analytical and design decisions were reviewed and validated by the author, who is responsible for the final outputs and conclusions.